

Problem Set 3

Due March 27, 2026 by 6pm

Please show all your work and include any code you create along with your submission. Be sure to acknowledge any substantial help you receive. This includes collaboration with others, references and online resources.

1. Using polynomial interpolation to approximate $\sqrt{2}$.

- (a) Determine the Lagrange form of the polynomial interpolant $p(x)$ for the given three $(x, f(x))$ data points $\{(-1, 1/2), (0, 1), (1, 2)\}$.
- (b) Simplify your polynomial interpolant. Noting that the data points lie on the graph of $f(x) = 2^x$, evaluate your polynomial interpolant at a suitable point to estimate $\sqrt{2}$.
- (c) Using known error estimates for polynomial interpolation, place a bound on the maximum error of your approximation to $\sqrt{2}$.

2. Approximating $\ln 2$ with piecewise polynomial interpolation.

- (a) How many coefficients are needed to define a piecewise *quadratic* interpolant of $n + 1$ data points? Recall that a piecewise quadratic interpolation of $n + 1$ data points (x_i, y_i) takes the form

$$s_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 \quad \text{for } x \in I_i = [x_i, x_{i+1}]$$

What conditions on s_i and derivatives should be imposed to determine these coefficients? Assume only one boundary condition.

- (b) Construct a piecewise quadratic interpolant of the data from the previous problem (reproduced below) using the boundary condition that the derivative vanishes at $x = -1$.

$$\{(-1, 1/2), (0, 1), (1, 2)\}$$

- (c) Use your piecewise polynomial interpolant to estimate the derivative $f'(x)$ at $x = 0$. Noting that the data points pass through the graph of $f(x) = 2^x$, explain why this should approximate $\ln 2$.

- (d) (Challenge) A better estimate can be obtained by using a different choice of boundary condition. For example, assuming that the second derivative is continuous at $x = 0$ leads to $\ln 2 \approx 0.75$. Can you think of a boundary condition that would give a better approximation *without* having to evaluate $\ln 2$ to implement it?

3. Consider the data points:

x	0	1	2
y	2	-1	4

- (a) Construct the Neville's table *by hand* for the given data set to approximate the value at $\bar{x} = 1.3$. Also *write code* that implements Neville's algorithm and verify your code produces the same approximation.
- (b) Carry out Newton's divided difference table *by hand* for the given data set to approximate the value at $\bar{x} = 1.3$. Also *write a code* that implements Newton's divided difference algorithm.
4. Consider interpolating $f(x) = xe^{-x}$ over $[-1, 3]$ with a polynomial of degree at most four. Interpolate at (a) uniformly spaced points, (b) the scaled and translated Chebyshev points and (c) the scaled and translated Legendre points. Calculate and compare the L_∞ norm and the L_2 norm of the interpolation error for all three polynomials, and comment on your results vs what would be predicted.

You may want use the divided difference algorithm from class to evaluate the polynomials over a grid of points on the given interval compute the error.

5. Write a script that computes the coefficients of a finite difference scheme for approximating the n th derivative of f at a point $\bar{x}$ using a given stencil of k grid points x_j .

$$f^{(n)}(\bar{x}) = \sum_{j=1}^k c_j f(x_j) + \text{Error}$$

The input should be in the form of (i) an integer n specifying the order of derivate, and (ii) an array of k values $\Delta x_j = x_j - \bar{x}$ specifying the relative location of the gridpoints. The output should be an array of coefficients c_j that minimize error of the approximate derivative. Use this script for the following problems.

- (a) Show that the script reproduces the known results for (i) forward first derivative, (ii) centered first derivative, and (iii) centered second derivative.
- (b) Use your code to show the expected convergence for approximated the first derivative of $f(x) = e^x$ at $x = 0$ using the centered finite difference approximation
- (c) Use your code to create a fourth-order forward finite difference approximation to the first derivative of $f(x) = \sin(x)$ at $x = 0$. Numerically verify that it is fourth order accurate.

6. Let $D = f'(x_0)$ denote the true derivative of the function $f(x)$ at x_0 and let

$$D_h = \frac{f(x_0) - f(x_0 - h)}{h}$$

denote the first-order backward difference approximation to the derivative.

- (a) Show that $D = D_h + k_1h + k_2h^2 + k_3h^3 + k_4h^4 + o(h^4)$ where k_i are constants independent of h .

Hint: Use Taylor's theorem to expand $f(x_0 - h)$ about the point $x = x_0$.

- (b) Explicitly compute the coefficients of the error formula for the approximation using $f(x) = \ln(x^2 + 1)$ and $x_0 = 1$.
- (c) Starting from $h = 1$, approximate the value of the first derivative of the given f at the given x_0 . By applying extrapolation to D_h . Use 4 rows in your extrapolation table.
- (d) What is the error of your final extrapolated approximation? What rate of convergence do you achieve? What step size would be needed in the first-order backward difference approximation to achieve the same error?